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Project Definition and Source Research

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Real Madrid Pulse: A Sentiment Analysis from Social Media and Sports News

0.1. Motivation

The analysis of public opinion in sports contexts has grown significantly with the rise of social media and digital journalism. Real Madrid, as one of the most followed football clubs in the world, generates a constant and massive flow of opinions, reactions, and discussions across multiple digital platforms. Understanding how fans and media perceive the team's performance throughout a season provides valuable insights that go beyond traditional statistics. Without this system, an analyst would need to manually browse Reddit threads and read dozens of news articles each day to build a picture of public sentiment a slow, subjective, and non-scalable process.

This project is motivated by the need to systematically capture and analyze this sentiment. By integrating both perspectives into the Real Madrid Pulse pipeline, the system automates ingestion, cleaning, and sentiment processing from both social and editorial sources. This approach makes it possible to identify emotional trends, detect key events that shift public opinion, and understand the relationship between match outcomes and fan sentiment, delivering a unified and quantified view of fan perception.

0.2. Goal

The goal of this project is to design and implement an end-to-end data engineering and NLP pipeline that collects, processes, and analyzes public opinion about Real Madrid from two complementary sources the Reddit API and the Football España website in order to deliver interactive sentiment insights through a visual dashboard, enabling a functional user to understand how public perception of the team evolves throughout the 2026 season in this case.

0.3. Scope and Limitations:

In scope:

Daily ingestion of posts and comments from Reddit (r/realmadrid, r/soccer) via the public Reddit API Daily web scraping of news articles from football-espana.net, Real Madrid section Text preprocessing and sentiment classification in English Pipeline implementation following Medallion architecture (Bronze → Silver → Gold) Orchestration with Apache Airflow and containerization with Docker Interactive dashboard built with Plotly Dash

Out of scope:

Paid or private data sources Real-time ingestion Official club data, match statistics, or player performance metrics Analysis of other football clubs or competitions beyond Real Madrid.

Known limitations:

Reddit API rate limits (60–100 requests/minute) may restrict data volume during high-activity periods User-generated content may contain sarcasm, irony, or informal language that reduces NLP accuracy Web scraping results depend on the website's HTML structure remaining stable

1. Functional User Definition:

The primary functional user of the Real Madrid Pulse dashboard is a Sports Data and Fan Sentiment Analyst — a professional responsible for monitoring, interpreting, and acting upon public opinion about Real Madrid across digital media channels. This role can be found either within the club's own digital and marketing department or in an external sports media outlet or digital agency working with the club.

Role: Sports Data and Fan Sentiment Analyst, operating in one of two contexts:

Internal context: Part of Real Madrid's marketing, communications, or fan engagement team, using sentiment data to evaluate the emotional impact of club decisions and design strategic responses. External context: Working for a sports media outlet or digital agency, using sentiment data to produce editorial content, audience reports, and coverage strategies around the club.

Technical level: Intermediate. The user is comfortable navigating interactive dashboards, interpreting NLP-derived metrics, and reading trend visualizations, but does not write code or manage data pipelines directly.

Information needs:

Track how fan sentiment evolves day by day, especially around match results, player signings, managerial decisions, and press statements. Distinguish between what sports media reports and what fans actually feel, identifying gaps or contradictions between both narratives. Detect early signals of opinion shifts both positive and negative before they escalate or fade.

Decision-making context:

From an internal perspective: The user decides when to launch fan engagement campaigns, how to frame official communications after a controversial event, which players to feature in marketing content based on positive fan association, and whether a recent club decision has damaged or strengthened the club's public image.

From an external perspective: The user decides what topics to cover editorially, when to publish opinion pieces, and how to summarize the club's public perception for stakeholders or brand partners.

2. User Stories:

The following user stories reflect both the internal (club/marketing) and external (media/agency) dimensions of the functional user, and are prioritized to guide dashboard design in later workshops.

Priority	**User Story**	**Benefit**
High	As a *Sports Data & Fan Sentiment Analyst*, I want to see a daily sentiment trend line for Real Madrid across both Reddit and news articles, so that I can continuously monitor how public perception evolves over time without manually reading every post or article.	Core insight delivery
High	As a *Club Marketing Strategist*, I want to measure how fan sentiment shifts before and after a specific club event — such as a match result, a player signing, or a press statement — so that I can evaluate the real emotional impact of that decision on the fan base and adjust future communications accordingly.	Strategic impact measurement
High	As a *Sports Data & Fan Sentiment Analyst*, I want to compare sentiment scores from news articles versus Reddit posts on the same day, so that I can detect whether media narratives align with or contradict what fans are actually feeling.	Cross-source analysis

Priority	User Story	Benefit
High	As a *Sports Data & Fan Sentiment Analyst*, I want to filter all sentiment views by a custom date range, so that I can focus my analysis on specific periods such as a match week, a transfer window, or a losing streak.	Temporal control
Medium	As a *Club Marketing Strategist*, I want to identify which players are most associated with positive sentiment in fan discussions, so that I can prioritize those players in marketing campaigns, sponsorship content, and social media features.	Brand & marketing strategy
Medium	As a *Sports Data & Fan Sentiment Analyst*, I want to view the most frequently mentioned keywords and named entities — players, opponents, and events — extracted from both sources in a given week, so that I can identify the dominant narratives driving public discourse around the team.	Topic discovery
Medium	As a *Club Fan Engagement Manager*, I want to be notified when overall fan sentiment drops significantly over consecutive days, so that I can trigger targeted engagement campaigns — such as exclusive content, fan events, or motivational messaging — to rebuild emotional connection with the supporter base before the situation worsens.	Proactive fan engagement
Medium	As a *Sports Data & Fan Sentiment Analyst*, I want to see which Reddit posts generated the highest engagement — measured by score and comment count — during a selected period, so that I can understand which topics resonated most strongly with the fan community and why.	Audience insight
Low	As a *Sports Data & Fan Sentiment Analyst*, I want to view a data quality summary for each ingestion cycle — including missing fields, duplicate records, and total records processed — so that I can trust that the insights displayed on the dashboard are based on clean and reliable data.	Data governance
Low	As a *Club Communications Analyst*, I want to export a structured sentiment summary report for a selected time period, so that I can share quantified findings with club directors, brand partners, or editorial managers without requiring them to access the dashboard directly.	Stakeholder reporting

Cuadro 1: User stories prioritized by importance for dashboard design.

3. API Source Research:

3.1. Selected API: Reddit API

For the development of this project, the Reddit API was selected as the main source of data. Reddit is an online discussion platform where users share opinions, comments, and reactions about different topics, including sports teams and football matches. This makes it a valuable source for sentiment analysis, as it provides real-time user-generated content from fans.

The Reddit API allows access to public posts and comments using HTTP requests and returns structured data in JSON format, which facilitates automated data collection and analysis.

3.2. API Documentation

Base URL:

`https://www.reddit.com/`

Authentication Method:

The public Reddit API allows basic data access without authentication, provided that a valid Üser-Agent"header is included in the request. This enables data retrieval in the free access level.

Available Endpoints:

- `/r/{subreddit}.json` – Retrieves posts from a specific subreddit
- `/search.json` – Searches posts using keywords
- `/comments/{post_id}.json` – Retrieves comments from a specific post

Example Endpoint Used in the Project:

`https://www.reddit.com/search.json?q=Real%20Madrid%20-Castilla&limit=25`

Rate Limits:

The Reddit API allows approximately 60 to 100 requests per minute for public access. Exceeding this limit may result in temporary blocking.

Data Format:

All responses from the API are returned in JSON format.

Relevant Fields for Analysis:

- title
- selftext
- author
- score
- num_comments
- created_utc
- subreddit

3.3. Justification for API Selection

The Reddit API is appropriate for this project because it provides direct access to opinions and discussions from football fans. Unlike traditional news sources, Reddit contains spontaneous user-generated content that reflects emotional reactions to team performance.

The textual fields such as post titles and comments will be used to perform sentiment analysis and evaluate public perception of the team's performance during the year 2026.

3.4. API Accessibility Verification

A test request was performed using Python to verify that the API is accessible and returns valid data.

```python
import requests
import json

url =  https://www.reddit.com/search.json

```

```python
headers = {
     User-Agent :  python:sentiment.analysis:v1.0 (by /u/test)
}

params = {
     q :  Real Madrid -Castilla ,
     limit : 20
}

response = requests.get(
    url,
    headers=headers,
    params=params,
    timeout=10
)

print( Status code: , response.status_code)

# Verificar si la respuesta es v lida
if response.status_code == 200:

    try:
        data = response.json()

        contador = 1

        for post in data[ data ][ children ]:

            info = post[ data ]

            print(
                contador,
                 | ,
                info.get( title ),
                 | ,
                info.get( author ),
                 | ,
                info.get( subreddit ),
                 | ,
                info.get( score ),
                 | ,
                info.get( num_comments )
            )

            contador += 1

    except json.JSONDecodeError:
        print( La respuesta no es JSON v lido )
        print(response.text)

else:

    print( Error en la solicitud )
    print(response.text)
```

Listing 1: API Request Test

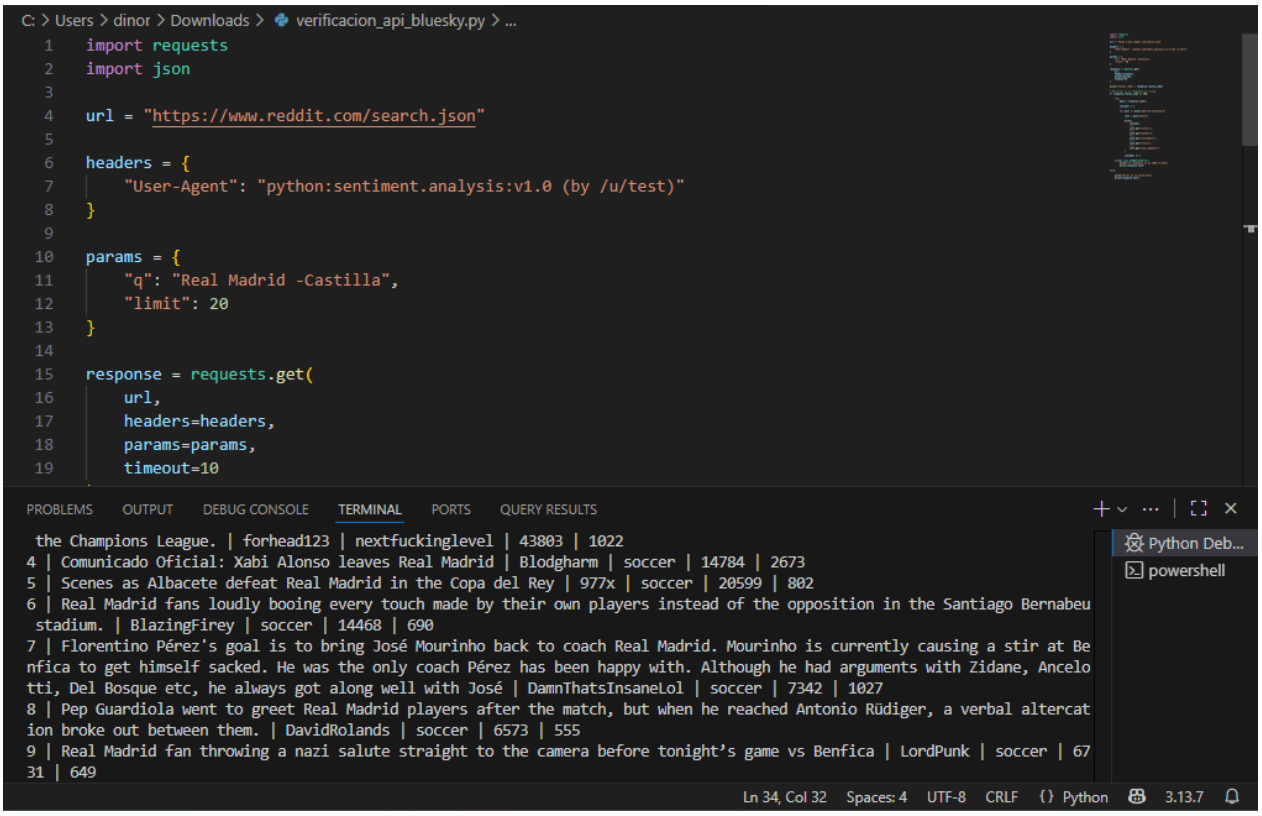

Figura 1: extracted from Visual Studio Code

4. Web Scraping Source Research:

4.1. Selected Data Source

To supplement the information obtained from APIs, the following website was selected as a data source:

- **Football España** – Real Madrid news section: `https://www.football-espana.net/category/la-liga/real-madrid`

This page contains news articles with text rich in analysis, opinions, and sports commentary, making it suitable for text analysis and natural language processing tasks.

4.2. Ethical and Legal Considerations

The website's `robots.txt` file was checked, yielding the following rules:

```
User-agent: *
Disallow: /wp-admin
Allow: /wp-admin/admin-ajax.php
Disallow: /wp-login
Disallow: /xmlrpc.php
Disallow: /wp-content/themes/*/$
Disallow: /trackback
Disallow: */trackback
```

Based on this configuration, we conclude that:

- There are no restrictions on access to public news pages.
- The blocked paths correspond to internal system functions.
- Scraping of content visible to the user is permitted.

4.3. Target Pages and HTML Structure

The scraping process is performed on individual article pages within the site.

Information is captured directly from multiple HTML tags to determine what valuable data can be extracted from the website.

The main tags analyzed include:

- Headings: `h1, h2, h3, h4, h5`
- Textual content: `p, span, div`
- Links: `a`
- Lists: `li`
- Emphasis: `strong, em`
- Dates: `time`

In addition, relevant attributes are extracted, such as:

- URLs (`href`)
- Images (`src`)
- CSS classes
- Document meta tags

4.4. Extracted Fields (Raw Bronze Data)

Unlike structured scraping, in this phase we opted for a **universal** approach, capturing as much information as possible.

This approach allows data to be stored in a **semi-structured** format, which is characteristic of the Bronze layer.

4.5. Rationale for Raw Data Approach

It was decided to capture the data comprehensively and without filtering because:

- In the Bronze phase, the priority is on capturing all the data
- It allows for flexibility in defining the schema in later stages
- It prevents the loss of potentially relevant information

Later, in the Silver layer, this data will be transformed into a cleaner, more analytical structure.

4.6. Complementarity with API Source (Reddit)

The API used in the project is **Reddit**, which provides user-generated data.

Both sources offer complementary perspectives:

- **Football España (Web Scraping):**
  - Professional editorial content
  - Structured information
  - Context and journalistic analysis
- **Reddit API:**
  - User opinions
  - Open discussion
  - Real-time subjective reactions

Integrating both allows us to bring fans' sentiments and opinions through the API and the team's context through daily team news

5. Data Sample Collection and Validation:

5.1. Data Collection API

A minimum of twenty (20) records were manually collected from the Reddit API using keyword-based searches related to the Real Madrid football team. The search query excluded the term Castilla to ensure that the collected data focused exclusively on the main professional team.

The selected records represent fan opinions, match reactions, and discussions about team performance. These records were extracted directly from the API response and stored in structured JSON format.

The collected data includes textual content suitable for sentiment analysis and Natural Language Processing (NLP) tasks.

5.1.1. Data Types Identification

The collected dataset contains different types of data, which are summarized in Table 2.

Cuadro 2: Data Types of Collected Fields

Field Name	Data Type	Description
id	String	Unique identifier of the post
title	String	Textual content of the post
author	String	Username of the author
subreddit	String	Community name
score	Integer	Number of upvotes
num_comments	Integer	Number of comments
created_utc	Integer	Unix timestamp of publication

5.1.2. Data Quality Evaluation

A basic data quality evaluation was performed on the collected samples to identify potential issues that could affect the reliability of the analysis.

The following observations were identified:

- Some records contain missing values in optional fields such as selftext.
- Certain posts include informal language, abbreviations, or slang commonly used in online communities.
- Some entries contain special characters or emojis, which may require encoding normalization.
- A small number of records may include irrelevant content unrelated to team performance.
- Duplicate posts may appear due to repeated discussions of the same match or event.

These issues are typical in social media datasets and require preprocessing before performing sentiment analysis.

5.2. Suitability for Sentiment Analysis and NLP

The collected dataset contains a significant amount of textual data in the form of user-generated content, including opinions, reactions, and discussions about team performance.

Each record includes a text field (title or selftext) that reflects the emotional state and perception of the fans. These textual elements provide sufficient information to perform sentiment analysis and Natural Language Processing (NLP) tasks.

Therefore, both the volume and the quality of the collected text data are considered adequate for:

- Sentiment classification
- Opinion mining
- Text preprocessing
- Natural Language Processing (NLP)

5.2.1. Preliminary Data Cleaning Requirements

Based on the evaluation of the collected data samples, several data cleaning steps were identified as necessary before performing sentiment analysis.

The required cleaning operations include:

- Removing duplicate records
- Converting text to lowercase
- Removing URLs and hyperlinks
- Eliminating special characters and punctuation
- Handling missing or empty text fields
- Removing stop words
- Normalizing encoding (UTF-8)
- Filtering irrelevant or off-topic posts

These cleaning steps will improve the quality and consistency of the dataset and ensure accurate Natural Language Processing (NLP) results.

5.2.2. Suitability for Sentiment Analysis and NLP

The collected dataset contains a significant amount of textual data in the form of user-generated content, including opinions, reactions, and discussions about team performance.

Each record includes a text field (title or selftext) that reflects the emotional state and perception of the fans. These textual elements provide sufficient information to perform sentiment analysis and Natural Language Processing (NLP) tasks.

Therefore, both the volume and the quality of the collected text data are considered adequate for:

- Sentiment classification
- Opinion mining
- Text preprocessing
- Natural Language Processing (NLP)

5.3. Data Collection Web Scraping

20 records were manually collected from the sports news website Football España, specifically from the Real Madrid section. Each record corresponds to an individual article, from which multiple structured and unstructured fields were extracted.

The main fields extracted are:

- **url**: article link
- **h1_all_texts**: main heading texts
- **h2_all_texts, h3_all_texts, h4_all_texts, h5_all_texts**: secondary headings
- **p_all_texts**: article text content (paragraphs)
- **span_all_texts, div_all_texts**: additional structural content
- **a_all_texts**: link text
- **li_all_texts**: list items
- **strong_all_texts, em_all_texts**: emphasized text
- **time_all_texts**: text associated with dates
- **all_hrefs**: links on the page
- **all_src**: image sources
- **all_classes**: CSS classes used on the page
- **all_meta**: document metadata
- **time_all**: complete structure of timestamp tags
- **author**: article author (when available)
- **published_time**: publication date extracted from meta tags

5.3.1. Data Quality Assessment web scraping

During the analysis of the collected data, several relevant quality issues were identified:

- **Missing fields**: Some articles do not contain explicit author information or a structured date.
- **Data noise**: irrelevant content was detected within the paragraphs, such as:
  - Comment sections ("Live Comments")
  - Form messages ("Leave a Reply")
  - Repeated navigation text
- **Redundant data**: multiple fields contain duplicate information (e.g., text in `div`, `span`, and `p` tags).
- **Inconsistent date formatting**: dates appear both in plain text and in `datetime` attributes.
- **Non-normalized structure**: data is stored as lists of text, requiring additional processing for analysis.
- **Encoding**: special characters (e.g., typographic quotes or UTF-8 symbols) were observed that require normalization.

5.3.2. Preliminary Data Cleaning Requirements

Based on the previous assessment, the following data cleaning requirements were identified:

- Filter only the relevant paragraphs from the main body of the article.
- Remove sections containing comments, forms, and navigation content.
- Normalize dates to a standard format (e.g., ISO 8601).
- Consolidate redundant fields into a single column of clean text.
- Clean up special characters and encoding issues.
- Standardize author names when available.

5.3.3. Suitability for NLP and Sentiment Analysis

The data obtained from scraping contains a significant amount of narrative and descriptive text, making it highly suitable for natural language processing (NLP) tasks, including:

- Sentiment analysis of sports news.
- Topic modeling.
- Entity detection (players, teams, events).

In particular, the `p_all_texts` field provides sufficient content in each article to effectively apply textual analysis models.

Therefore, it is concluded that the source is suitable and complementary for the development of the proposed analysis pipeline.

5.4. Sample Data Records API

Table 3 presents a subset of collected records used for initial validation and analysis.

Cuadro 3: Sample Data Retrieved from Reddit API

ID	Post Title	Author	Score	Comments	Date	Subreddit
1	Benfica [4] - 2 Real Madrid - Anatoliy Trubin 90+8’ (Goalkeeper)	ayoefico	33537	2273	1769637795	soccer
2	Real Madrid [3] - 0 Manchester City - Federico Valverde 42’	ayoefico	11249	1387	1773261831	soccer
3	Benfica goalkeeper Anatoliy Trubin scores a last minute goal vs Real Madrid to save his team from being eliminated from the Champions League.	forhead123	43803	1022	1769662881	nextfuckinglevel
4	Comunicado Oficial: Xabi Alonso leaves Real Madrid	Blodgharm	14790	2673	1768238045	soccer

5.5. Sample Data Records Web Scraping

The following is a simplified example of a record in tabular format:

	url	h1_all_texts	h2_all_texts	h3_all_texts	h4_all_texts	h5_all_texts	p_all_texts	span_all_texts	div_all_texts	a_all_texts	li_all_texts	strong_all_texts	em_all_texts	time_all_texts	all_hrefs	all_src	all_classes	all_meta	time_all	author	published_time
0	https://www.football-espana.net/2026/03/21/rud...	['Antonio Rudiger is an example for young play...	['Rudiger is an example for young players', '11...	[Live Comments, Leave a ReplyCancel reply]	[]	[]	[Real Madrid manager Alvaro Arbeloa has defend...	[Tags, Add a Comment, Your email address will ...	[HomeLatest NewsAnalysisLa LigaAlavesAthletic ...	[Home, Latest News, Analysis, La Liga, Alaves, ...	[Home, Latest News, Analysis, La LigaAlavesAth...	[]	[]	[21 March 2026, 17:48, 21 March 2026, 17:48, 2...	[https://www.football-espana.net/, /, /categor...	[https://fcdn.football-espana.net/wp-content/u...	[a2a_dd, form-submit, std-pad, icon, boxout, s...	{'viewport': 'width=device-width, initial-scal...	[{'text': '21 March 2026, 17:48', 'datetime': ...	Ruairidh Barlow	21 March 2026, 17:48
1	https://www.football-espana.net/2026/03/21/alv...	[Alvaro Arbeloa explains how Jude Bellingham w...	[Arbeloa: Bellingham return won't affect Thiag...	[1 Comment, Leave a ReplyCancel reply]	[]	[]	[Real Madrid are back in La Liga action on Sun...	["Tomorrow is a very important match for us be...	[HomeLatest NewsAnalysisLa LigaAlavesAthletic ...	[Home, Latest News, Analysis, La Liga, Alaves, ...	[Home, Latest News, Analysis, La LigaAlavesAth...	[]	[]	[21 March 2026, 17:03, 21 March 2026, 17:03, 2...	[https://www.football-espana.net/, /, /categor...	[https://fcdn.football-espana.net/wp-content/u...	[a2a_dd, form-submit, std-pad, icon, boxout, c...	{'viewport': 'width=device-width, initial-scal...	[{'text': '21 March 2026, 17:03', 'datetime': ...	John Menzies	21 March 2026, 17:03
2	https://www.football-espana.net/2026/03/21/sim...	[Diego Simeone praises Álvaro Arbeloa before M...	[Changes to Real Madrid since first meeting, W...	['Everyone has their ups and downs' – Simeone ...	[]	[]	[Atlético Madrid manager Diego Simeone has bee...	[Tags, Add a Comment, Your email address will ...	[HomeLatest NewsAnalysisLa LigaAlavesAthletic ...	[Home, Latest News, Analysis, La Liga, Alaves, ...	[Home, Latest News, Analysis, La LigaAlavesAth...	[]	[]	[21 March 2026, 15:35, 21 March 2026, 15:35, 2...	[https://www.football-espana.net/, /, /categor...	[https://fcdn.football-espana.net/wp-content/u...	[a2a_dd, form-submit, std-pad, icon, boxout, s...	{'viewport': 'width=device-width, initial-scal...	[{'text': '21 March 2026, 15:35', 'datetime': ...	Ruairidh Barlow	21 March 2026, 15:35
3	https://www.football-espana.net/2026/03/21/rea...	[Real Madrid defender brings forward injury re...	[Real Madrid will ease Militao back in]	[Live Comments, Leave a ReplyCancel reply]	[]	[]	[Real Madrid have dealt with many injury probl...	[Tags, Add a Comment, Your email address will ...	[HomeLatest NewsAnalysisLa LigaAlavesAthletic ...	[Home, Latest News, Analysis, La Liga, Alaves, ...	[Home, Latest News, Analysis, La LigaAlavesAth...	[]	[]	[21 March 2026, 14:42, 21 March 2026, 14:42, 2...	[https://www.football-espana.net/, /, /categor...	[https://fcdn.football-espana.net/wp-content/u...	[a2a_dd, wp-image-339156, form-submit, std-pad...	{'viewport': 'width=device-width, initial-scal...	[{'text': '21 March 2026, 14:42', 'datetime': ...	John Menzies	21 March 2026, 14:42
4	https://www.football-espana.net/2026/03/21/man...	[Manchester United target prioritising summer ...	[Is Guimaraes the answer to Real Madrid's midf...	[1 Comment, Leave a ReplyCancel reply]	[]	[]	[Real Madrid are planning to sign a new midfie...	[Tags, Add a Comment, says: Your email address...	[HomeLatest NewsAnalysisLa LigaAlavesAthletic ...	[Home, Latest News, Analysis, La Liga, Alaves, ...	[Home, Latest News, Analysis, La LigaAlavesAth...	[]	[]	[21 March 2026, 13:19, 21 March 2026, 13:19, 2...	[https://www.football-espana.net/, /, /categor...	[https://fcdn.football-espana.net/wp-content/u...	[a2a_dd, form-submit, std-pad, icon, boxout, c...	{'viewport': 'width=device-width, initial-scal...	[{'text': '21 March 2026, 13:19', 'datetime': ...	John Menzies	21 March 2026, 13:19

Figura 2: extracted table from web scraping

6. Extraction Strategy Definition:

6.0.1. Ingestion Periodicity

The data collection frequency from the sources is set to daily.

Since the analyzed website constantly publishes news related to Real Madrid, especially during the sports season, it is necessary to maintain a regular collection schedule that allows for capturing new articles without overloading the server. Similarly, the Reddit API for the r/realmadrid subreddit generates continuous daily activity with discussions, reactions to matches, and performance analyses, even on days without official matches.

A daily frequency allows for both sources:

- To capture new articles and comments published at regular intervals.
- To keep the information up to date for subsequent sentiment analysis in near real time.
- To avoid excessive requests that could affect the site’s availability or reach Reddit API rate limits.

- Synchronize both sources (web + social media) within the same ingestion cycle for precise temporal correlations.

This daily frequency is optimal considering:

- **Web scraping**: 3–10 daily news articles about Real Madrid depending on news activity.
- **Reddit API**: 20–100 daily post on r/realmadrid.

6.0.2. Raw Data Format

The extracted data is stored in **CSV** format, allowing for a simple tabular representation that is easy to process using tools such as Pandas.

Each generated file follows a timestamp-based naming convention:

`football_espana_real_madrid_YYYYMMDD_HHMMSS.csv`

This approach allows for:

- Maintaining a history of executions (data versioning).
- Preventing previously collected information from being overwritten.
- Facilitating auditing and traceability of the ingestion process.

6.0.3. Metadata Fields

Metadata consists of attributes that describe each record and allow the collected data to be contextualized. In this scraping process, the main metadata identified are:

- **url**: link to the original article.
- **author**: author of the article (when available).
- **published_time**: publication date of the article.
- **source**: data source (Football España).
- **category**: content category (Real Madrid / La Liga).
- **ingestion_timestamp**: date and time the data was extracted.

This metadata allows us to identify the origin of the data, organize it correctly, and maintain traceability within the pipeline.

6.0.4. Justification of Periodicity

The decision to use a daily frequency is based on the fast-paced nature of the sports world and the ongoing nature of discussions on social media.

Both the analyzed website (football-espana.net) and the Reddit API (r/realmadrid) constantly generate content related to Real Madrid:

A daily data ingestion allows for:

- Capturing new content published regularly on both sources.

- Keeping information up-to-date for timely sentiment analysis.
- Synchronizing web and social media data for temporal correlations (e.g., news + Reddit reaction).

This frequency is optimal given the observed activity:

- **football-espana.net**: Continuous news, especially during the high season.
- **r/realmadrid**: Daily engagement (average of 27K weekly contributions).

7. Pipeline Architecture Overview:

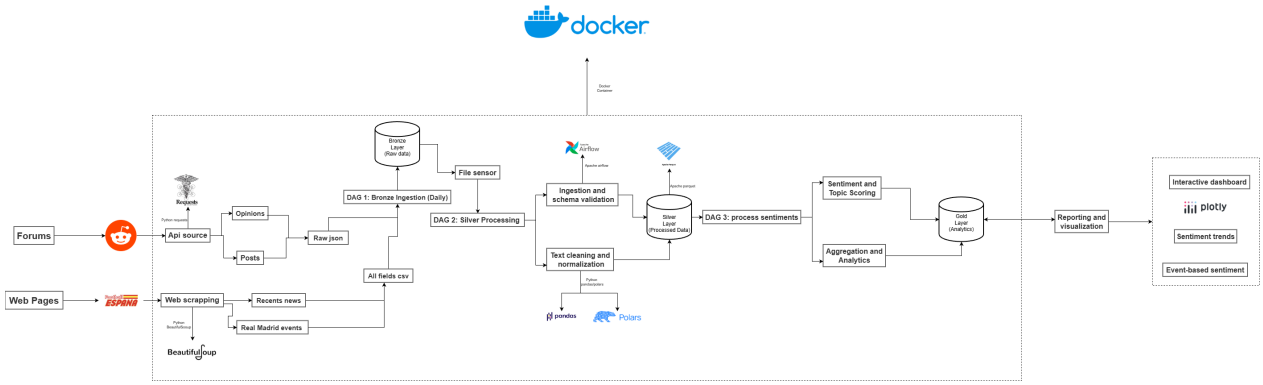

Figura 3: High-level diagram of the proposed data pipeline

The proposed pipeline follows a Medallion architecture (Bronze → Silver → Gold), which allows data to be organized into different levels of processing, quality, and analytical value. This structure facilitates data traceability, from its raw form through to its use in visualizations and advanced analyses.

7.1. Bronze Layer (Raw Data)

The **Bronze** layer corresponds to the raw data ingestion stage. In this phase, data is collected directly from the original sources and stored in a format that is as close as possible to the original, without any significant transformations.

The main sources of data are:

- Web scraping of sports websites
- external apis

Tools such as Python are used for data extraction, along with libraries like `requests` and `BeautifulSoup`. This layer is orchestrated using an Apache Airflow DAG, scheduled to run daily.

Data is stored in JSON format for the API and in CSV format for data retrieved from the website within the `datalake_bronze/` directory. This layer retains unprocessed information, which may include noise, duplicates, or irrelevant data, but ensures full traceability of the process.

7.2. Silver Layer (Processed Data)

The Silver layer is responsible for transforming raw data into structured, cleaned data that is ready for analysis.

The following processes are carried out at this stage:

- Data schema validation
- Text cleaning (removal of HTML, unnecessary characters)
- Field normalization (dates, authors, content)
- Duplicate removal

Tools such as Python are used for these transformations, along with libraries like Pandas and Polars. The orchestration is handled by a second DAG in Apache Airflow, which is automatically triggered by a FileSensor. This sensor detects the arrival of new files in the Bronze layer and triggers processing in Silver.

The processed data is stored in Parquet format in the `datalake_silver/` directory, enabling more efficient and optimized access for subsequent analysis.

7.3. Gold Layer (Analytics and Aggregation)

The Gold layer contains data ready for analytical use. At this stage, aggregations, metrics, and advanced analyses are generated.

The main processes include:

- Sentiment analysis of news articles and comments
- Topic modeling
- Aggregations such as:
  - Number of news articles by date
  - Sentiment trends
  - Identification of relevant events

The technologies used include Python, Pandas, and PySpark. Orchestration is performed using a third DAG in Apache Airflow.

The results are stored in the `datalake_gold/` directory, in Parquet formats optimized for analytical use.

7.4. Visualization Layer

The visualization layer allows users to explore processed data through interactive dashboards.

Plotly Dash is used to build interactive web applications that enable users to:

- View sentiment trends
- Analyze events related to Real Madrid
- Explore news and aggregated metrics

This layer directly consumes data from the Gold layer.

7.5. Orchestration Strategy

The pipeline is orchestrated using Apache Airflow, which employs multiple DAGs to automate and coordinate each stage of the process:

- **DAG 1: Bronze Ingestion (Daily)**
  Runs daily to extract data from APIs and via web scraping, storing the results in JSON format in the Bronze layer.
- **DAG 2: Silver Processing**
  Triggered by a FileSensor that detects new files in Bronze. Performs data cleaning, validation, and transformation, generating files in Parquet format.
- **DAG 3: Gold Processing**
  Processes data from the Silver layer, performs sentiment analysis, and generates aggregations for analysis.

7.6. Infrastructure

To ensure the system's reproducibility and portability, **Docker** is used to containerize the various components of the pipeline, including:

- Apache Airflow
- Scraping and processing scripts
- Visualization dashboard

8. GitHub repository

All materials related to Workshop 1, including reports and diagrams are available in the following GitHub repository:

- Programming for Data Analysis course repository link

References

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